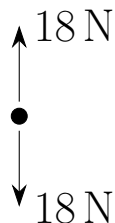
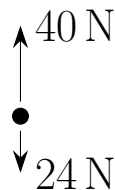

Task #1

In each of the free-body diagrams below, calculate the **magnitude** and **direction** of the net force.

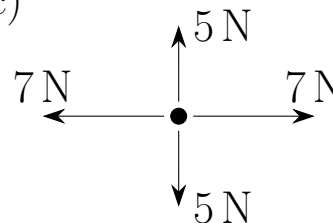
(a)



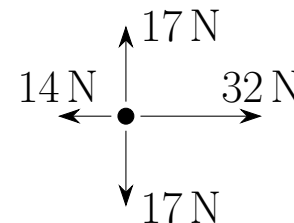
(b)



(c)



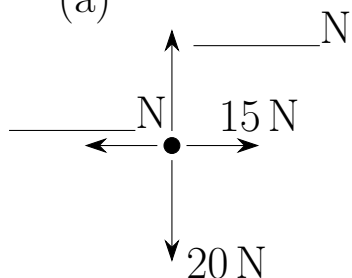
(d)



Task #2

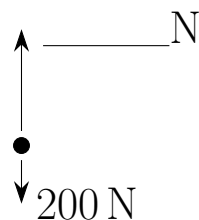
In each of the free-body diagrams below, the net force is given, but one or more of the applied forces is missing. Find the missing forces.

(a)



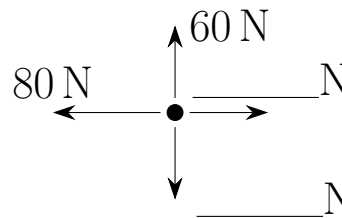
$$F_{NET} = 0 \text{ N}$$

(b)



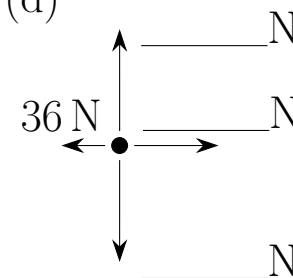
$$F_{NET} = 150 \text{ N, up}$$

(c)



$$F_{NET} = 45 \text{ N, left}$$

(d)

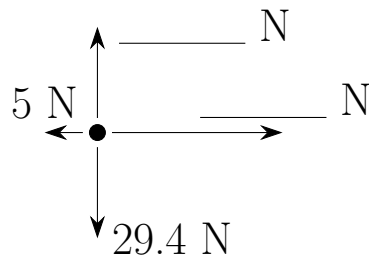


$$F_{NET} = 23 \text{ N, right}$$

Task #3

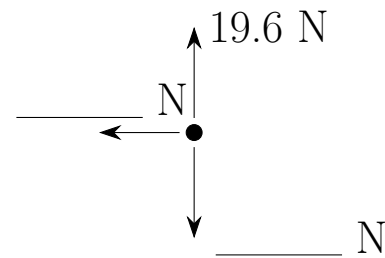
Fill in the blanks in each of the situations depicted below. Draw the net force.

(a)



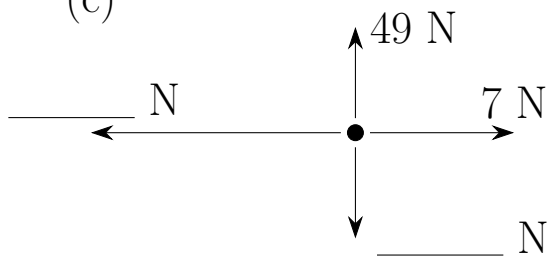
$$\begin{aligned} m &= 3 \text{ kg} \\ a &= \text{____ m/s}^2, \text{ ____} \\ F_{NET} &= 23 \text{ N, right} \end{aligned}$$

(b)



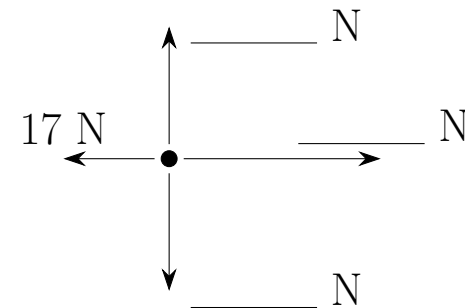
$$\begin{aligned} m &= 2 \text{ kg} \\ a &= 8 \text{ m/s}^2, \text{ left} \\ F_{NET} &= \text{____ N, ____} \end{aligned}$$

(c)



$$\begin{aligned} m &= 5 \text{ kg} \\ a &= 12 \text{ m/s}^2, \text{ left} \\ F_{NET} &= \text{____ N, ____} \end{aligned}$$

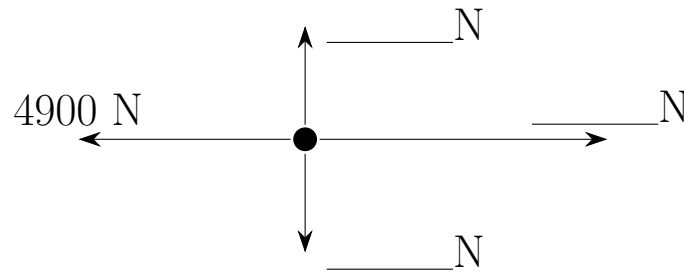
(d)



$$\begin{aligned} m &= 3 \text{ kg} \\ a &= 18 \text{ m/s}^2, \text{ right} \\ F_{NET} &= \text{____ N, ____} \end{aligned}$$

Task #4

A 900-kg car accelerates from rest to 45 m/s in 6.8 seconds. Fill in the free-body diagram below, assuming that the average force against the motion of the car due to air drag and friction is 4900 N.



Task #5

Consider 15,000-kg rocket that is accelerating upward at 18.2 m/s^2 . Fill in the forces on the free-body diagram below.

